HomeOps User & Evaluation Guide

Autonomous Smart Home Operations Agent for Amazon Alexa+ & AWS Bedrock

1. Executive Overview

Welcome to **HomeOps**, built for the **Build, Ship, Shape: Amazon Developer Hackathon 2026** (Alexa+ Track, AWS Builder Challenge, and Open Source Mini Challenge). HomeOps is a next-generation autonomous smart home operations agent. Unlike traditional voice assistants that require rigid, single-turn commands ('Alexa, turn on the porch light'), HomeOps uses a multi-step **ReAct (Reasoning + Acting)** framework powered by **AWS Bedrock** to orchestrate interdependent home subsystems (HVAC, solar storage, smart deadbolts, HEPA filtration, and freeze-sensitive plumbing) based on high-level human directives.

Component	Technology / Service	Key Role in HomeOps
Alexa+ Agent	AWS Bedrock (Claude 3.5 / Nova)	Multi-step ReAct reasoning, tool selection & voice synthesis
Smart Mesh	Virtual Physical Node Bus	Real-time telemetry, compressor power draw, sensor simulation
Cloud State	Amazon DynamoDB	Audit logging, state persistence, plan history storage
Proactive Bus	AWS EventBridge	Ambient trigger ingestion (tariff spikes, freeze warnings)
Open Source SDK	@homeops/alexa-plus-sdk	Reusable TypeScript device adapters and safety guardrails

2. Quickstart: Launching the App in Under 2 Minutes

Both backend and frontend services can run simultaneously on your local machine:

```
Terminal 1 (Backend API & WebSocket Server):
  cd backend
  npm run dev     (Runs on http://localhost:4000)

Terminal 2 (Interactive Alexa+ Web Console):
  cd frontend
  npm run dev     (Runs on http://localhost:3000)
```

Open your web browser and navigate to: **http://localhost:3000**

3. Visual Tour of the HomeOps Interface

The web console is divided into three functional areas:

• **Top Navigation & Scenario Bar:** Contains quick status indicators (AWS Builder, Open Source SDK, Live WebSocket) and the 4 interactive scenario injection buttons.

• **Left Column — Alexa+ Multimodal Console & ReAct Stream:** Echo Show 15 simulation featuring the pulsing Alexa cyan acoustic ring, speech-to-text mic input, speech synthesis audio playback, and live ReAct step-by-step reasoning logs.

• **Right Column — Smart Home Device Mesh:** Live tiles representing the 8 home subsystems: Living Room Thermostat, Master Bedroom Thermostat, Front Door Smart Deadbolt, Back Patio Sliding Lock, Solar & Battery Reserve, HEPA Air Purifier Pro, Water Main Sentinel, and Ring Perimeter Floodlight.

4. Step-by-Step Evaluation Scenarios

Use these 4 pre-configured test scenarios to evaluate how HomeOps reasons autonomously:

Scenario A: Bedtime Security Lockdown

1. **Setup:** Click **'Night Security Vulnerability'** in the top bar (sets patio door ajar).
2. **Trigger:** Speak or click: *'Alexa, prepare the house for bedtime'*.
3. **Autonomous Result:** Agent sweeps perimeter → locks front deadbolt → locks sliding patio door → arms Ring floodlights → lowers bedroom temperature to 68°F. Spoken confirmation confirms full night mode.

Scenario B: Utility Peak Tariff Shaving

1. **Setup:** Click **'Peak Tariff Spike ($0.48/kWh)'** in the top bar.
2. **Trigger:** Speak or click: *'Alexa, electricity rates are high, save me some money'*.
3. **Autonomous Result:** Agent reads tariff surge → engages battery discharge to buffer house load → sets HVAC compressors to Eco mode → cuts grid power draw by 62%.

Scenario C: Cold Snap Freeze Protection

1. **Setup:** Click **'Freezing Pipe Warning (Basement 31°F)'** in the top bar.
2. **Trigger:** Speak or click: *'Alexa, freeze alert incoming, protect my water pipes'*.
3. **Autonomous Result:** Agent scans water line sensors → detects 31°F risk → boosts zone heating to 68°F → safely raises pipe temperature to 46°F to prevent pipe burst.

Scenario D: Smoke & Air Quality Purge

1. **Setup:** Click **'Smoke / Poor Air Quality'** in the top bar (simulates AQI 184).
2. **Trigger:** Speak or click: *'Alexa, I smell smoke in the living room'*.
3. **Autonomous Result:** Agent detects dangerous particulate spike → sets HEPA filter to Turbo speed → rapidly cleans air back to safe levels (AQI 32).

5. Interactive Manual Actuation (Bidirectional Sync)

Every device tile in the mesh includes interactive manual controls to test two-way synchronization. When an action is taken manually, changes are instantly transmitted via WebSockets to all connected clients:

Subsystem	Interactive Controls Available	Physics & Telemetry Effect
Thermostats	[+] / [-] temp dials, [Switch Eco]	Updates target temp; Eco drops draw from 410W to 130W
Smart Locks	[Lock Deadbolt], [Toggle Ajar]	Toggles locked state; ajar/unlocked triggers security alert
Energy Storage	[Peak Surge Tier] toggle	Switches pricing tier between standard and peak surge rate
Air Purifier	[Turbo Fan] / [Auto Fan]	Turbo increases fan RPM, clears airborne particulate
Water Main	[Shut Valve] / [Open Valve]	Controls water isolation valve; prevents catastrophic leaks
Ring Security	[Armed], [■ Light On] toggle	Toggles exterior perimeter motion sensors and floodlights

6. Voice Microphone & Audio Synthesis

• **Real Microphone Input:** Click the microphone icon in the left search bar. Speak into your computer's microphone using natural language (e.g., *'Alexa, set temperature to 69 degrees'*). HomeOps uses the Web Speech Recognition API to transcribe your speech and dispatch the directive automatically.
• **Audio Voice Output:** Whenever Alexa+ replies, click the speaker icon (■) in the spoken response bubble to hear Alexa's voice synthesized aloud via the browser's SpeechSynthesis engine.

7. AWS Builder Cloud Stack & Open Source SDK

• **AWS CloudFormation / SAM Template:** Located at infra/template.yaml, defining the AWS Bedrock agent role, Lambda Action Group dispatchers, DynamoDB tables (HomeOps_Devices, HomeOps_Plans), and custom EventBridge bus (HomeOps-HomeEvents-Bus).
• **Open-Source SDK:** Located at sdk/, published as @homeops/alexa-plus-sdk under Apache-2.0. Contains plug-and-play smart device adapters, deterministic cyber-physical guardrails, and Bedrock action group schemas.

8. Troubleshooting & FAQ

Question / Issue	Explanation & Solution
Browser asks for microphone permission?	Click 'Allow' when clicking the microphone button so the Web Speech API can capture your audio.
What if AWS credentials are not set?	HomeOps has an intelligent built-in autonomous ReAct emulation engine with real telemetry so all features work 100% offline out-of-the-box.
How to reset the simulation?	Click the **'Reset System'** button in the top right of the scenario bar to restore all devices to normal baseline.

Submission Notice: *Prepared for the Amazon Developer Hackathon 2026.*
All source code is licensed under Apache-2.0. Video demo script is located in docs/DEMO_VIDEO_SCRIPT.md, and developer platform feedback is in docs/PRODUCT_FEEDBACK.md.